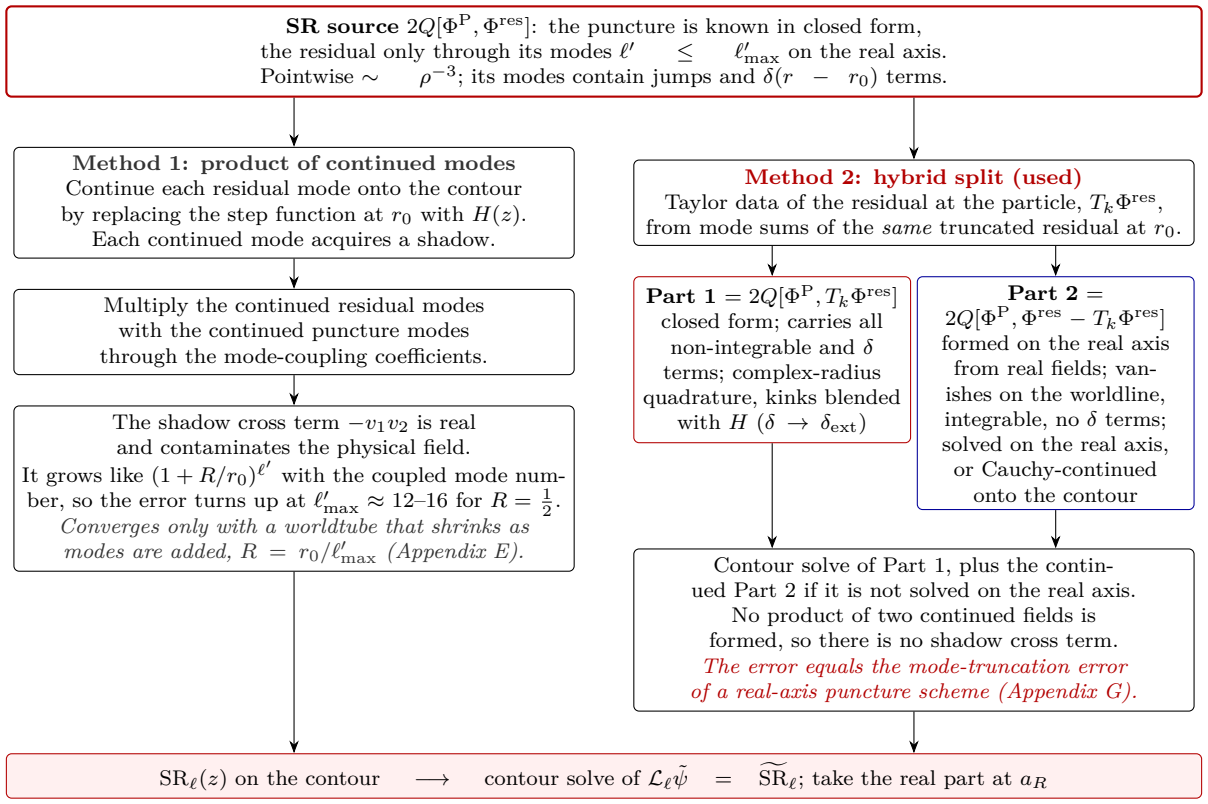

SR source $2Q[\Phi^{\rm P}, \Phi^{\rm res}]$: the puncture is known in closed form,
the residual only through its modes $\ell' \quad \leq \quad \ell'_{\rm max}$ on the real axis.
Pointwise $\sim \quad \rho^{-3}$; its modes contain jumps and $\delta(r \;-\; r_0)$ terms.
Method 1: product of continued modes
Continue each residual mode onto the contour
by replacing the step function at r_0 with $H(z)$.
Each continued mode acquires a shadow.
Multiply the continued residual modes
with the continued puncture modes
through the mode-coupling coefficients.
The shadow cross term $-v_1 v_2$ is real
and contaminates the physical field.
It grows like $(1 + R/r_0)^{\ell'}$ with the coupled mode number, so the error turns up at $\ell'_{\rm max} \approx 12$–$16$ for $R = \frac{1}{2}$.
Converges only with a worldtube that shrinks as modes are added, $R = r_0/\ell'_{\rm max}$ (Appendix E).
Method 2: hybrid split (used)
Taylor data of the residual at the particle, $T_k\Phi^{\rm res}$,
from mode sums of the *same* truncated residual at r_0.
Part 1 $= 2Q[\Phi^{\rm P}, T_k\Phi^{\rm res}]$
closed form; carries all non-integrable and δ terms; complex-radius quadrature, kinks blended with H $(\delta \;\to\; \delta_{\rm ext})$
Part 2 $=$
$2Q[\Phi^{\rm P}, \Phi^{\rm res} - T_k\Phi^{\rm res}]$
formed on the real axis from real fields; vanishes on the worldline, integrable, no δ terms; solved on the real axis, or Cauchy-continued onto the contour
Contour solve of Part 1, plus the continued Part 2 if it is not solved on the real axis.
No product of two continued fields is formed, so there is no shadow cross term.
The error equals the mode-truncation error of a real-axis puncture scheme (Appendix G).
$\mathrm{SR}_\ell(z)$ on the contour $\quad\longrightarrow\quad$ contour solve of $\mathcal{L}_\ell\tilde{\psi} \quad = \quad \widetilde{\mathrm{SR}}_\ell$; take the real part at a_R